

CodeRank

An Online Place to Practice Coding — Project Documentation

1. What is CodeRank?

CodeRank is a website where people can practice coding. It works like LeetCode or HackerRank. A user can sign up, pick a problem, write code in the browser, run it on sample tests, and then submit it to check against hidden tests. The user can see if the code is correct, see past submissions, and track their progress. An admin can add new problems and test cases. Later, we can add a leaderboard and user profiles.

2. Tools We Will Use

Part	Tool	Why we chose it
Backend	Java 21 + Spring Boot	Stable and easy to use for servers.
Frontend	React 19 + Vite	Fast to build modern web pages.
Database	MySQL 8.4	Reliable and well supported for a long time.
Login	JWT + Spring Security	Simple and safe way to handle logins.
Database code	Spring Data JPA	Makes saving and reading data easy.
Code editor	Monaco Editor	Same editor that VS Code uses.
Containers	Docker + Compose	Runs code safely and the same on any machine.
API docs	Springdoc OpenAPI	Auto-creates a page that shows all APIs.

Tip: Use the latest stable version of each tool when you start.

3. Features

We will build the basic features first (MVP). After that, we will add more features.

Module	Build first (MVP)	Add later
Login	Sign up, log in, user roles	Google login, email check, reset password
Problems	Show list, details, tags, levels	Company lists, topic sheets, paid problems
Judge	Run and check Java code	Add Python, C++, JavaScript; faster workers
Submissions	Show result, time, history	Compare two codes, find copied code
Admin	Add and edit problems & tests	Upload many at once, write editorials
Fun stuff	Simple leaderboard	Streaks, badges, ratings, contests

4. How the System Works

At first, keep it simple: one backend that does everything. Later, when things are working, move the code-checking part to its own service.

Normal request:

Browser → React (frontend) → Spring Boot (backend) → MySQL

When user submits code:

Backend → Judge Service → Docker Sandbox → Result (Pass / Fail)

Project folders in production:

- **frontend/** — the React website
- **backend/** — the Spring Boot server
- **judge-worker/** — runs user code safely
- **mysql/** — stores all data
- **nginx/** — sends traffic to the right place and adds HTTPS